Shockingly Simple Monetary Shocks

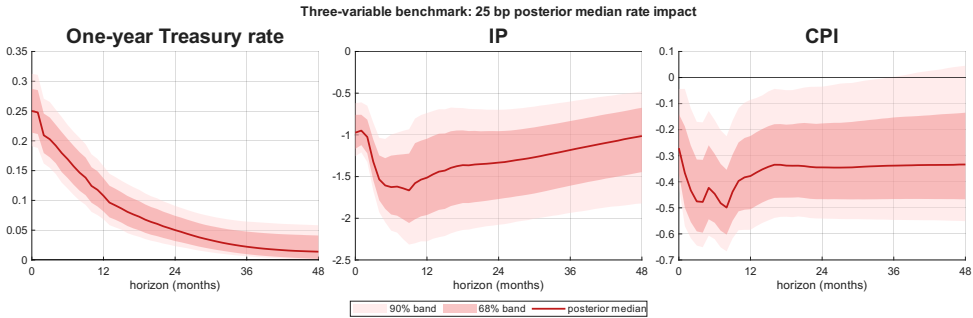
Tom D. Holden Mu-Chun Wang

Deutsche Bundesbank

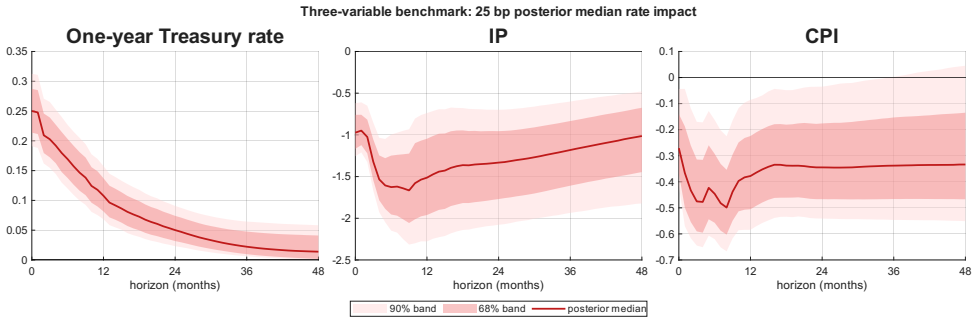
2026-09-21

The views expressed in this paper are those of the author and do not represent the views of the Deutsche Bundesbank, the Eurosystem or its staff.

What kind of shock is this? Rate ↑, Output ↓, Prices ↓



What kind of shock is this? Rate ↑, Output ↓, Prices ↓



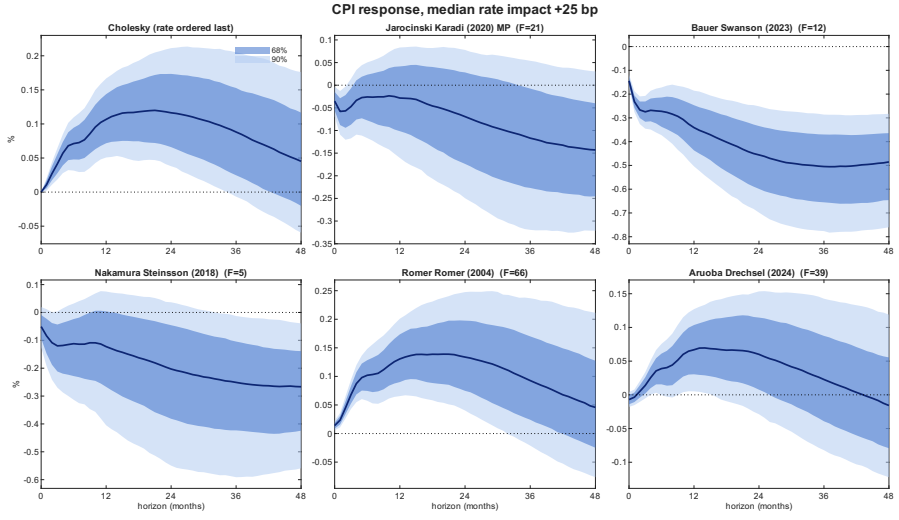
Produced with two simple restrictions:
Rates are positive and decline monotonically following monetary shocks.

How the profession identifies monetary shocks

- **High-frequency**: Gertler Karadi (2015), Nakamura Steinsson (2018), Jarociński Karadi (2020), Miranda-Agrippino Ricco (2021), Bauer Swanson (2023).
- **Narrative**: Romer Romer (2004).
- **Text / NLP**: Aruoba Drechsel (2024).

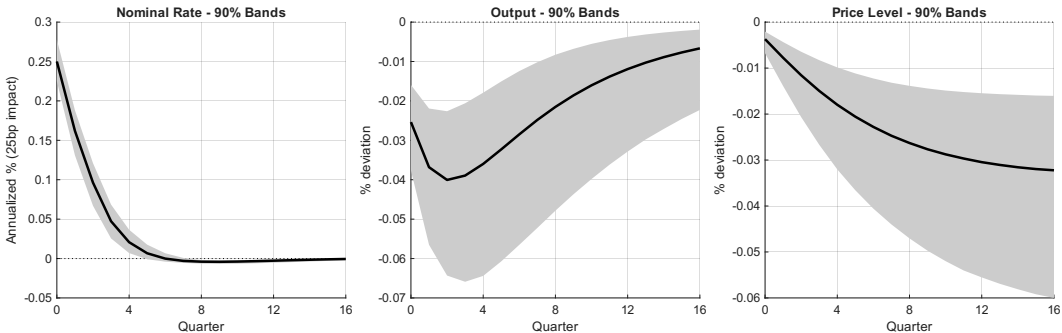
- Powerful: But only where a valid, strong instrument exists.

Instrument-based VARs disagree on the price response



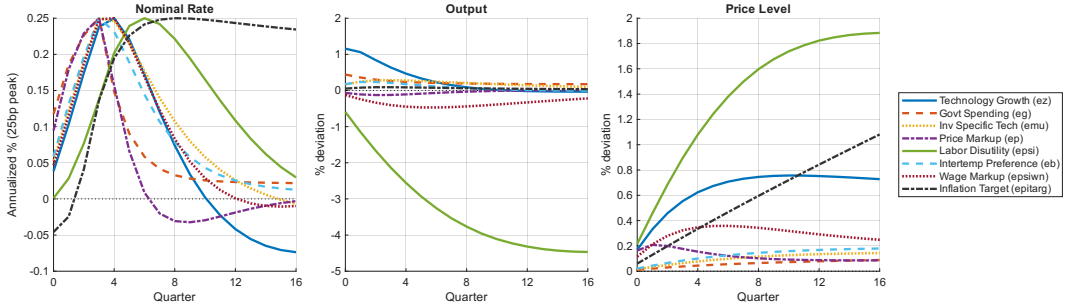
Four-variable VAR with EBP, 1973–2019. Cholesky and Romer–Romer show a CPI rise; several instruments show a fall. **Median** rate impact: $+25$ bp; NS remains weak ($F \approx 5.2$).

How does the policy rate respond to monetary shocks in estimated medium-scale DSGE models?



IRFs to a monetary shock in Justiniano, Primiceri, and Tambalotti (JPT) (2013).
(Minuscule overshoot (UB: -0.2 b.p.).)

How does the policy rate respond to other shocks in estimated medium-scale DSGE models?

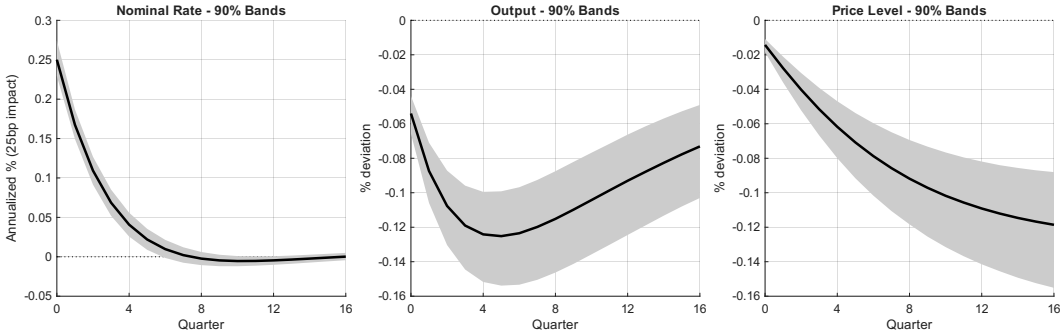


IRFs to non-monetary shocks in JPT (2013).
Non-monetary shocks generate hump-shaped rate dynamics.

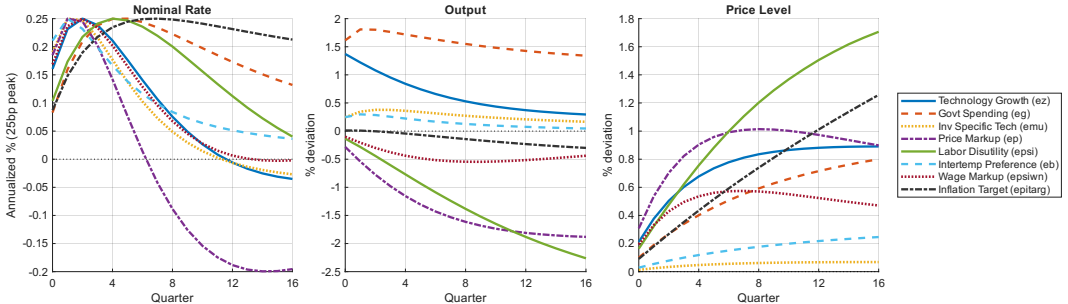
Improvements to the JPT model on an extended dataset

Specification / Extension	Log Marginal Data Density	
	Laplace	MHM
Baseline JPT on extended sample (1964:Q2–2019:Q4)	−1324.13	−1323.71
+ Correct how inflation target enters monetary rule	−1323.70	−1323.59
+ AR(2) government spending, one root at 0.995	−1313.39	−1313.03
+ AR(2) labor disutility, one root at 0.995	−1307.84	−1307.44
+ No price indexation (so price dispersion enters)	−1295.75	−1295.33
+ No wage indexation (so wage dispersion enters)	−1279.47	−1279.34
+ Contemporaneous Taylor rule	−1267.83	−1267.58
+ Endogenous fiscal response to output growth	−1263.90	−1263.74
+ Fix price markup persistence at 0.995	−1257.32	−1257.21
+ Fix technology growth persistence at 0	−1253.77	−1253.59
+ Monetary response to government spending growth	−1250.70	−1250.47

The improved JPT model removes the overshoot



Other shocks still have humped shape responses in the improved model



What is the optimal response to a monetary policy shock?

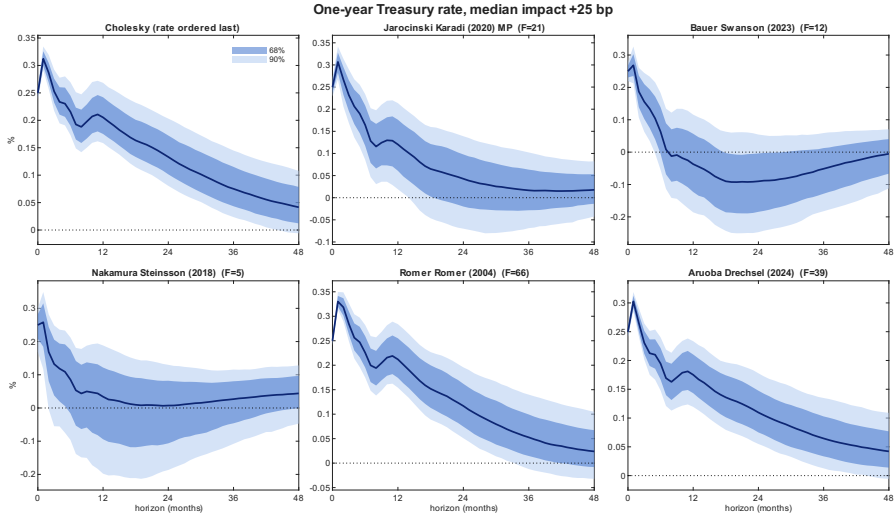
Model a monetary shock as one-period rise in the central bank's distaste for inflation. Objective: $\mu\pi_0 + \frac{1}{2}\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left(\pi_t^2 + \lambda x_t^2\right)$. Rest of model: Basic NK.

- To cut today's inflation the bank optimally uses both tools: Lower expected inflation and open an output gap.
- It commits to a persistent disinflation and recession.

- Nominal rates = real rates ($\uparrow$) + expected inflation ($\downarrow$).
- Under plausible calibrations, the real rate rise dominates.
- $\Rightarrow$ Nominal rates rise. One (co-)state $\Rightarrow$ monotonic decline.

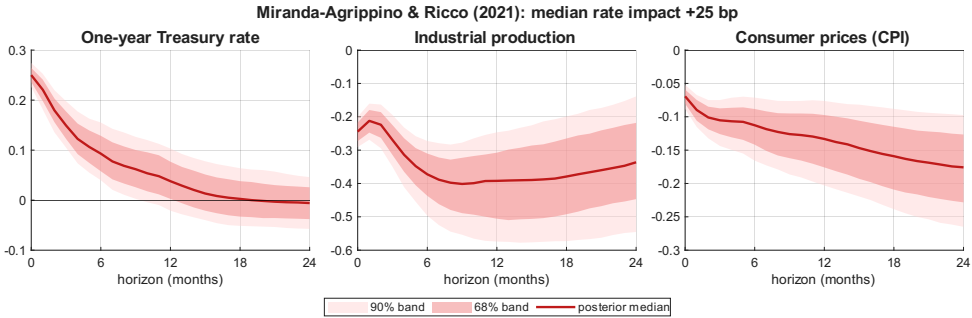
So, the optimal response to a monetary shock features a nominal interest rate that: **Rises on impact**, **declines monotonically** (no hump), **stays positive** (no overshoot).

Rate responses still show humps and overshooting



The same VAR: every median rate impact is $+25$ bp, but individual impacts vary.
Several rate paths have early peaks or humps; Bauer–Swanson's median overshoots below zero.

What a well-identified shock looks like



Miranda-Agrippino & Ricco's (2021) information-robust instrument:
The rate decays toward zero, output contracts, and prices fall without a price puzzle.

Their Figure 3 specification; 68% and 90% posterior bands; **median** rate impact +25 bp.
1979:1–2014:12. First-stage F-stat $F = 15.3$, reliability $\Lambda = 0.26$.

Our restriction: Monotone decay + non-negativity

Let q be the first column of the rotation matrix.
We can find vectors $a_0, a_1, \ldots$ such that the nominal rate IRF at horizon h is $a_h^\top q$.
Then, given a fixed cutoff h^*, we impose:

(i)	$a_0^\top q \geq 0$	rises on impact
(ii)	$-(a_h - a_{h-1})^\top q \geq 0$	$h = 1, \ldots, h^*$
(iii)	$a_{h^*}^\top q \geq 0$	non-negative at h^*

- Agnostic about decay *speed*, *magnitude*, *every other variable*.
- Linear in q.
- $\Rightarrow$ The admissible rotations form a convex spherical polytope.

Our prior and sampler

We use the following prior:

$$\pi(B, \Sigma) \propto \mathrm{NIW}_\lambda(B, \Sigma) \times \mathbf{1}[\{q \mid C_{B,\Sigma} q \geq 0, \|q\| = 1\} \neq \emptyset].$$

where $C_{B,\Sigma}$ collects the restrictions from the previous slide,
and NIW_λ uses the Giannone, Lenza, Primiceri (2015) version of the Minnesota prior.

We sample from (B, Σ) using a preconditioned Crank–Nicolson proposal.

Having sampled (B, Σ) from the posterior, we draw q uniformly from:

$$\{q \mid C_{B,\Sigma} q \geq 0, \|q\| = 1\}.$$

Difference to Arias et al. (2018)

- Arias et al. (2018) implicitly imposes a volume-weighted prior on (B, Σ).
- Values of (B, Σ) for which the identified set has small measure get downweighted.

- Our conditionally-uniform prior minimises distortion of reduced form posterior.
- Partial answer to Baumeister and Hamilton (2015).

> *"we were able to conduct inference based on a reference prior that is conditionally uniformly distributed on the (compact) identified set $\Theta(\phi)$. Such a prior might have appeal to a broad audience."*
>
> — *Moon and Schorfheide (2012)*

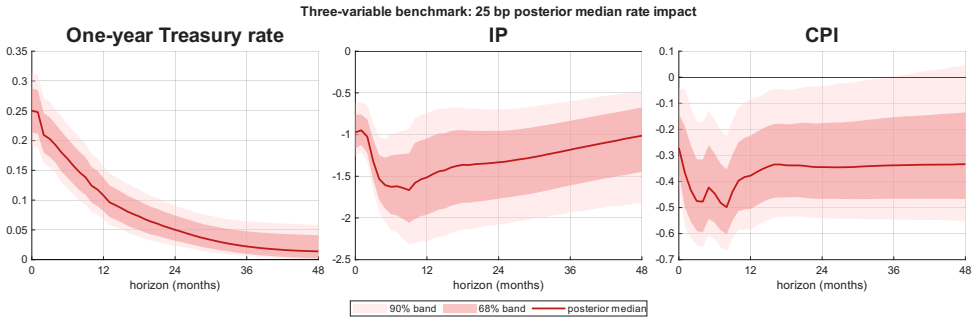
- Slight caveat: We are uniform in "q-space", not in IRF space.

Data and sample

- Three monthly U.S. variables:
  1. one-year Treasury yield (%);
  2. industrial production ($100 \times \log$);
  3. consumer price index ($100 \times \log$).
- Sample **1982:01–2019:12** (post-Volcker), **12 lags** ($T = 444$).

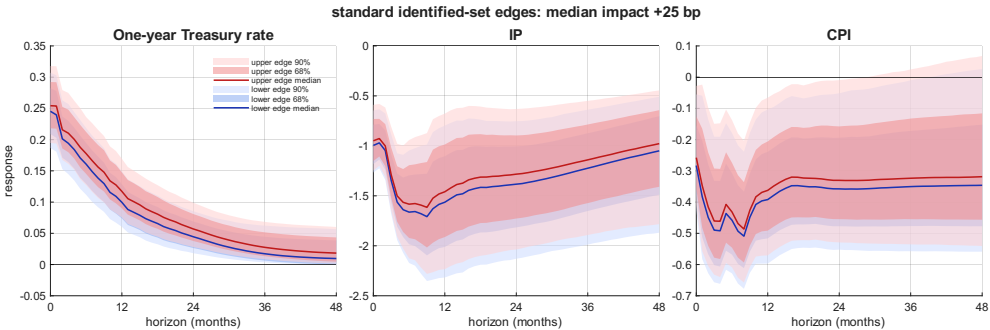
- **Loose** Minnesota prior, $\lambda = 5$ (far above the usual ≈ 0.2).
- Restriction horizon $h^* = 48$ months, imposed on the one-year yield.

Main result: A textbook tightening



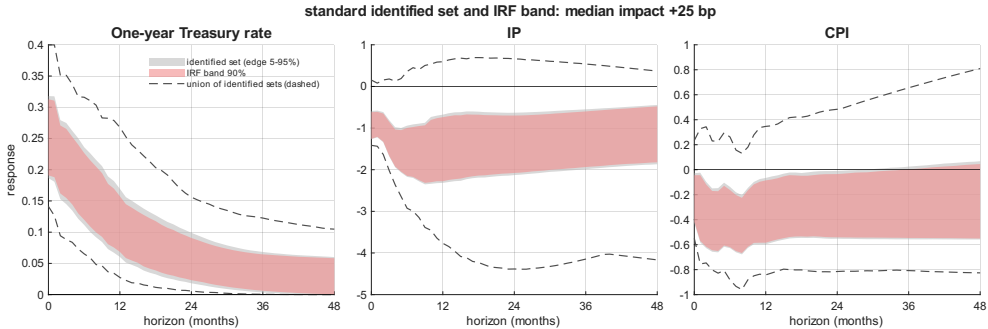
Rate rises and decays with no overshoot. Output contracts to a trough at $\sim$1 year.
The price level falls rapidly and permanently with no price puzzle.
No external instrument needed!

Edges of the identified set



Credible bands for the lower (blue) and upper (red) edges of the identified set.
Almost all the width reflects reduced-form uncertainty about where the set sits.
Near point identification! (Note: Edges are exactly point identified.)

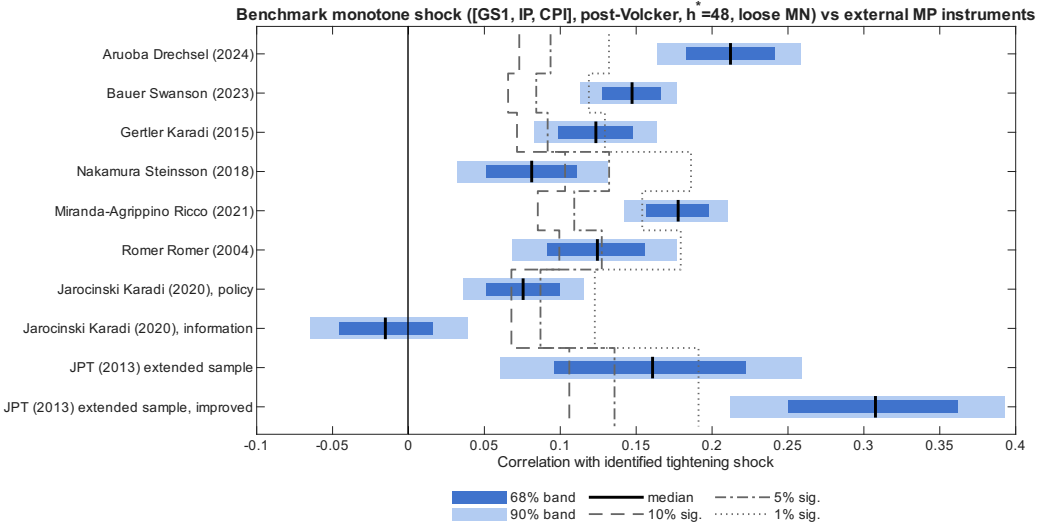
Union of identified sets



The dashed lines show the union of identified sets across all draws. Unsurprisingly wide.

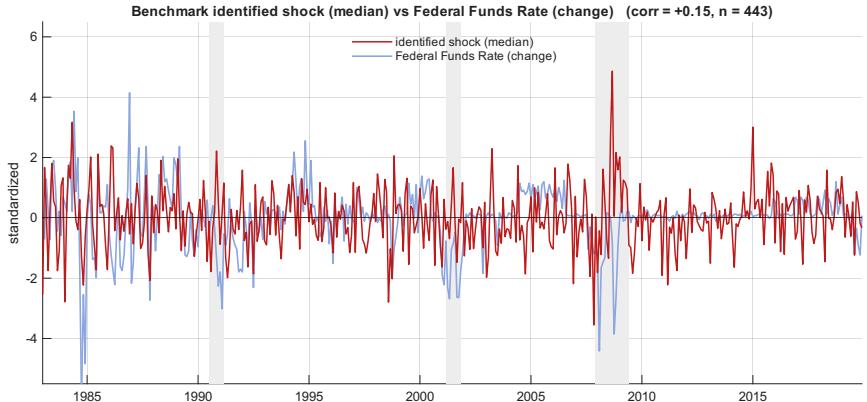
The grey area repeats the shaded area from the last slide.
The red area repeats the 90% credible bands from the main IRFs.
$\Rightarrow$ Baseline credible bands $\approx$ Credible bands of the upper & lower edges.

Validation versus other instruments

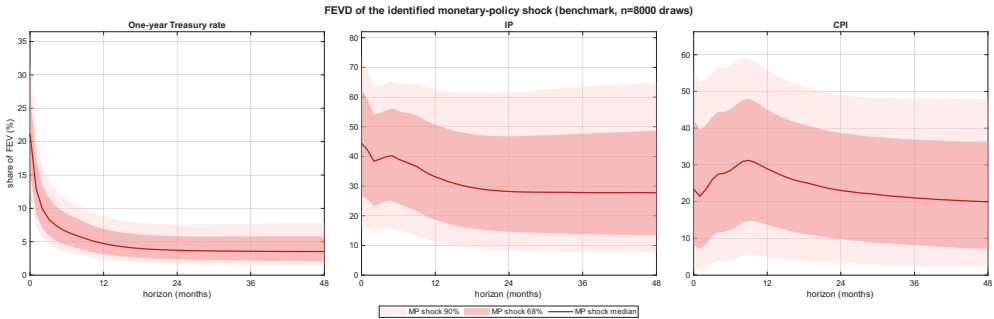


Correlates positively with all nine instruments. ≈ 0 corr with CB *information* shock, as desired.

The shocks

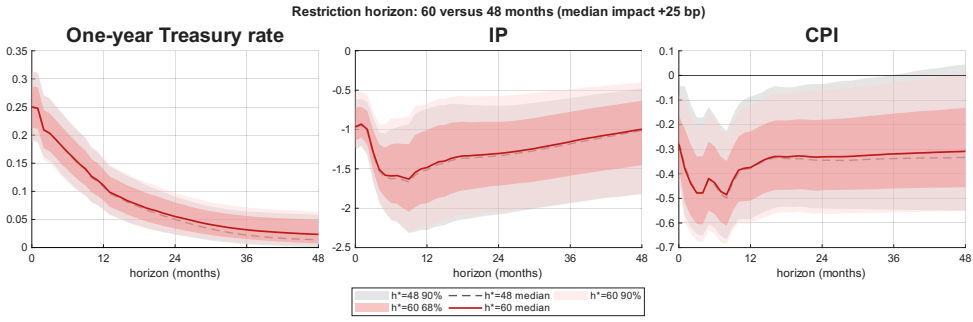

Benchmark identified shock (median) vs Federal Funds Rate (change) (corr = +0.15, n = 443)
identified shock (median)
Federal Funds Rate (change)
standardized
6
4
2
0
-2
-4
1985
1990
1995
2000
2005
2010
2015

Forecast error variance decomposition



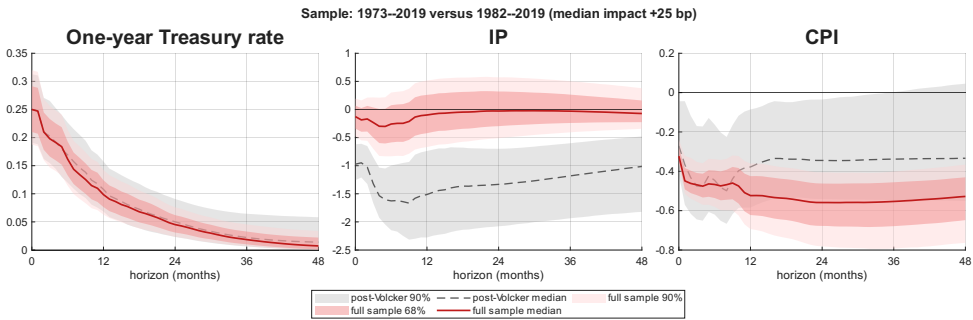
Share of forecast error variance explained by the identified monetary shock. Rather high, but this is inevitable in a small VAR. 68% and 90% credible bands.

Robustness: Restriction horizon



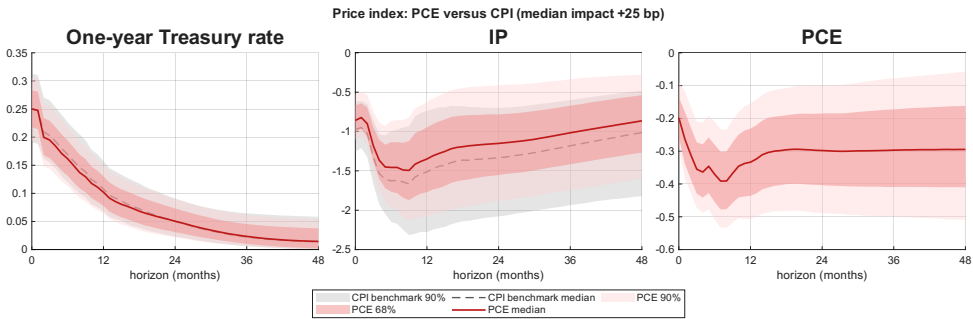
Benchmark $h^* = 48$ (grey) vs $h^* = 60$ (red) — the picture is unchanged.

Robustness: Sample period



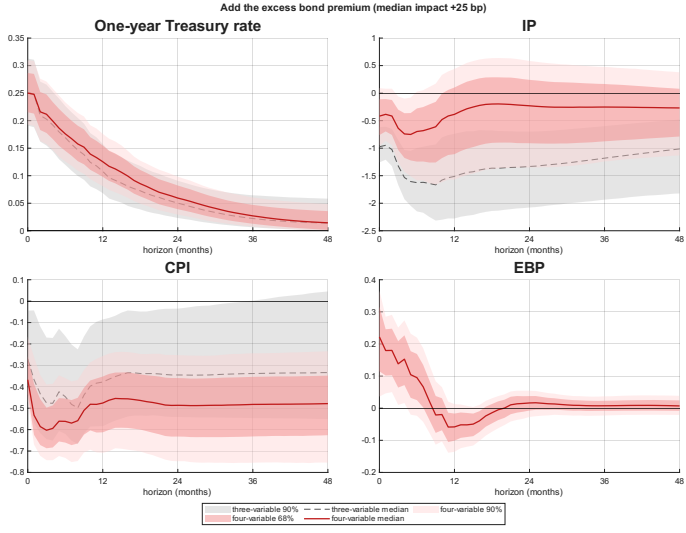
Post-Volcker benchmark 1982–2019 (grey) vs full sample 1973–2019 (red): Similar rate and price dynamics. Weaker output response.

Robustness: Price index



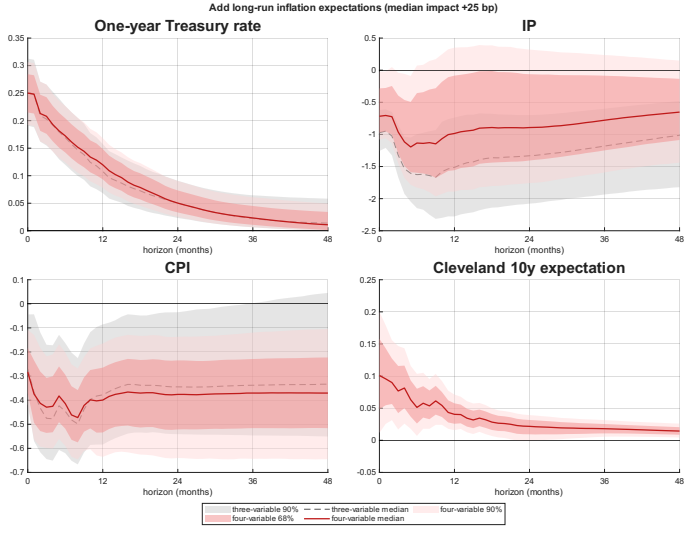
CPI benchmark (grey) vs the PCE price index (red) — no price puzzle under either.

Robustness: Adding the excess bond premium



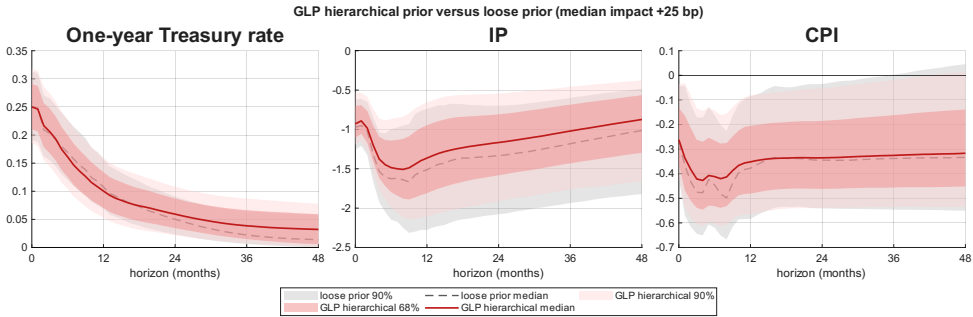
Adding the excess bond premium (red) vs. the three-variable benchmark (grey).

Robustness: Adding long run inflation expectations



Adding the Cleveland 10y expected-inflation series (red) vs the 3-var benchmark (grey).

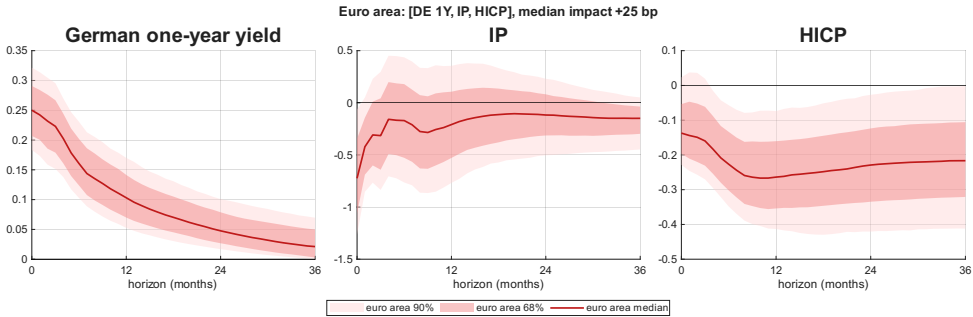
Robustness: Prior tightness



The Giannone, Lenza, Primiceri (2015) hierarchical prior $\bar{\lambda} = 0.34$ (red) vs...

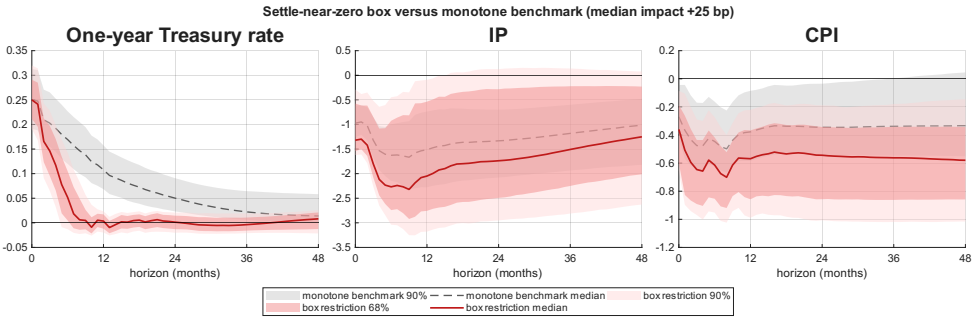
the loose benchmark $\lambda = 5$ (grey).

Robustness: The euro area



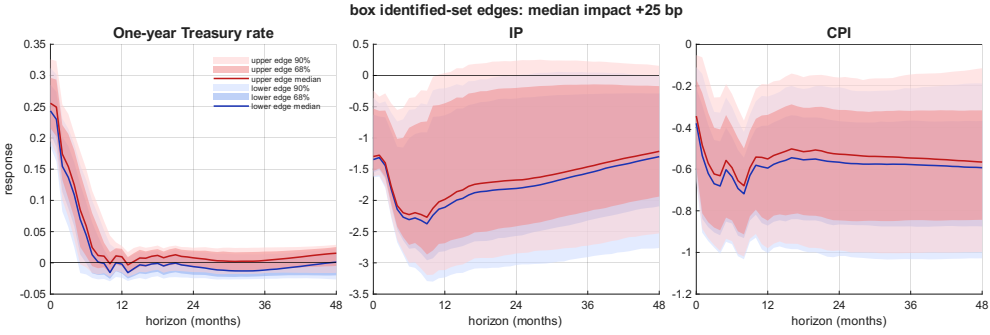
Euro area [German 1y yield, IP, HICP], 1999–2016, $h^* = 36$, same shape restriction.
Classic monetary tightening: HICP falls (no price puzzle) and output contracts.

Alternative “box” identification: Decline then settle near zero



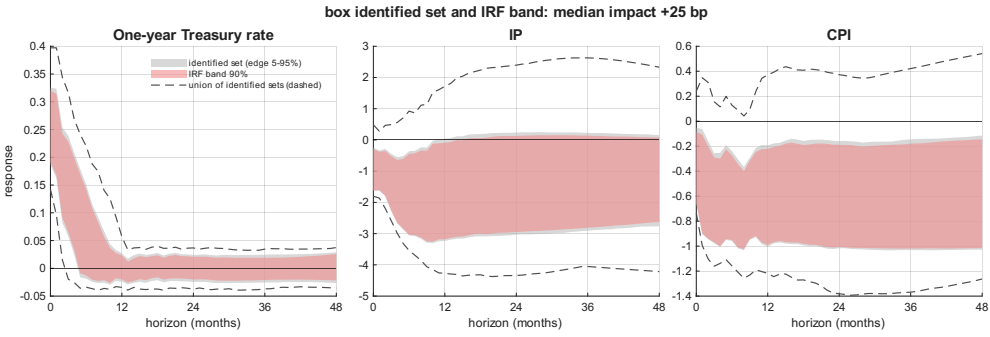
New constraints: Decay to zero, then stay within ± 2.5 bp (red).
The variable responses closely track the benchmark ones (grey).

Alternative “box” identification: Identified set edges



Credible bands (68% and 90%) for the upper and lower edges of the identified set.

Box identification: Identified-set envelope



IRF 90% band (red) vs. identified set (grey) vs. union envelope (dashed).

Conclusion

- Simple shape restrictions on policy-rate IRF: non-negative, monotonic decline.
- Delivers a textbook monetary tightening in a minimal three-variable VAR.

- No high-freq. info, no sign restriction on other variables, no recursive ordering.
- Correlates with all seven monetary-policy instruments, *not* information shock.

- Robust to horizon, sample, price index, prior, expectations, settle-near-zero box.

- Rapid response of prices to monetary shocks is not an artefact of HF identification.

Comments very welcome!

Appendix: What +25 bp means in these results

Let $R_i^{(d)}(h)$ be the unnormalised response in posterior draw d. We use one positive factor for all draws and all variables:

$$s = \frac{0.25}{\text{median}_d \, R_{\text{GS1}}^{(d)}(0)}, \qquad \widetilde{R}_i^{(d)}(h) = s R_i^{(d)}(h).$$

- The median rate impact is +25 bp; individual impacts still vary.
- We calculate medians and credible bands *after* applying this common factor.
- Since $s > 0$, normalisation cannot reverse a draw's sign.
- Each alternative specification receives its own median-impact scale.

Appendix: The pCN-MH sampler

Two block updates target $\pi(B,\Sigma) \propto \mathrm{NIW}_\lambda(B,\Sigma)\,\mathbf{1}[\mathcal{S}(B,\Sigma)\neq\varnothing]$ $\big(L=\mathrm{chol}\,\Sigma$, $R=\mathrm{chol}\,(X'X)^{-1}\big)$:

B — **preconditioned Crank–Nicolson proposal:**

$$B' = \bar{B} + \sqrt{1-\beta^2}\,(B-\bar{B}) + \beta\, R\, Z\, L', \qquad Z_{ij} \overset{\text{iid}}{\sim} \mathcal{N}(0,1).$$

It is reversible w.r.t. the matrix-normal prior $\mathcal{MN}\big(\bar{B}, \Sigma \otimes (X'X)^{-1}\big)$, so the Metropolis ratio collapses to accept iff $\mathcal{S}(B',\Sigma)\neq\varnothing$ — a pure feasibility check, with *no* posterior density to evaluate. ($\beta = 0.10$, $\approx$45% accepted.)

Σ — **Cholesky-symmetric random walk** (every 5th step):

$$L' = L\,\exp\big(\tfrac{\eta}{2}(W+W')\big), \qquad \Sigma' = L'L'^{\top}.$$

The proposal is symmetric in the affine-invariant metric; the acceptance probability combines the NIW Metropolis ratio and a Jacobian $(n+1)\eta\,\mathrm{tr}(W)$. ($\eta = 0.02$, $\approx$30%.)

Initialised at a feasible (B_0,Σ_0) found by rejection at a tight $\lambda_{\text{init}} = 0.1$, then run at the loose $\lambda = 5$.

Appendix: Drawing the rotation: Geodesic hit-and-run

Each feasible (B, Σ) the pCN-MH chain retains still needs a rotation q (first column of Q): the shock's impact vector is Lq and the rate response at horizon h is $a_h^\top q$. The monotone-decay restriction is the spherical polytope $\mathcal{S} = \{q \in \mathbb{S}^{n-1} : Cq \geq 0\}$.

We draw q uniformly from $\mathcal{S}$ by geodesic hit-and-run:

1. at the current q, pick a random tangent direction d;
2. move along the great circle $\gamma(s) = \cos(s)\, q + \sin(s)\, d$;
3. each row $c_k^\top \gamma(s) \geq 0$ defines a feasible arc; intersect the K arcs;
4. draw s uniformly on the feasible interval, set $q \leftarrow \gamma(s)$, and repeat.

The polytope is geodesically convex, so the feasible set on each great circle is a single arc — the chain is well defined and converges to the uniform law on $\mathcal{S}$. We take one q per retained (B, Σ) (burn-in 200, thin 10); only inequalities are imposed, so every draw enters with *equal weight*.